

Question 1

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The following equation relates the numbers r and s , where $s \neq 0$ and $r < 9$:

$$r = 9 - \frac{27}{s}.$$

Which equation expresses s in terms of r ?

a	$s = \frac{r+9}{27}$
b	$s = \frac{r-9}{27}$
c	$s = -\frac{27}{r-9}$
d	$s = \frac{27}{r-9}$
e	$s = \frac{9r-27}{27}$

EXPLANATION

We are given the equation

$$r = 9 - \frac{27}{s}$$

and asked to solve for s in terms of r .

First, we subtract 9 from *both sides*:

$$\begin{aligned} r - 9 &= 9 - \frac{27}{s} - 9 \\ r - 9 &= -\frac{27}{s} \end{aligned}$$

Next, we multiply *both sides* by -1 :

$$-(r - 9) = \frac{27}{s}$$

Then multiply *both sides* by s :

$$-(r - 9)s = 27$$

Finally, divide *both sides* by $-(r - 9)$ to isolate s :

$$\frac{-(r-9)s}{-(r-9)} = \frac{27}{-(r-9)}$$

$$s = -\frac{27}{r-9}$$

Therefore, the correct answer is

$$s = -\frac{27}{r-9}.$$

Question 2

?

The variables m and n are related by the equation

$$m = \frac{5n+6}{n}.$$

Which equation expresses n in terms of m ?

a	$n = \frac{5}{m-6}$
b	$n = \frac{6}{m-5}$
c	$n = \frac{1}{m-6}$
d	$n = \frac{5n}{m-6}$
e	$n = \frac{5m}{n+6}$

EXPLANATION

Multiplying both sides of the given equation by n gives

$$mn = 5n + 6.$$

Subtracting $5n$ from both sides, we get

$$mn - 5n = 6.$$

Factoring out n on the left-hand side, we get

$$n(m-5) = 6.$$

Finally, dividing both sides by $m-5$ yields

$$n = \frac{6}{m-5}.$$

So, the correct answer is

$$n = \frac{6}{m - 5}.$$

Question 3

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The variables z and q are related by the equation

$$z = \frac{6q + 12}{q}.$$

Which equation expresses q in terms of z ?

a	$q = \frac{6z}{q-12}$
b	$q = \frac{6}{z+12}$
c	$q = \frac{6q}{z+12}$
d	$q = \frac{1}{z-6}$
e	$q = \frac{12}{z-6}$

EXPLANATION

Multiplying both sides of the given equation by q gives

$$zq = 6q + 12.$$

Subtracting $6q$ from both sides, we get

$$zq - 6q = 12.$$

Factoring out q on the left-hand side, we get

$$q(z - 6) = 12.$$

Finally, dividing both sides by $z - 6$ yields

$$q = \frac{12}{z - 6}.$$

So, the correct answer is

$$q = \frac{12}{z - 6}.$$

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